

MOHAMMAD AZLAAN

AI Systems Engineer | Agentic Pipelines | GPU Inference

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PROFESSIONAL SUMMARY

Highly motivated AI systems engineer with expertise in LLM orchestration, agentic pipelines, and full-stack AI applications, seeking to leverage my skills in high-performance computing and graphics processing to drive growth and reduce dependence on key customers. With experience in developing innovative AI solutions, including a 4-stage LangGraph biomedical research agent and a fully local AI assistant on private GPU, I am confident in my ability to contribute to the development of high-performance computing and graphics processing technologies. I am excited about the opportunity to join Advanced Micro Devices and utilize my skills to drive growth and innovation in the semiconductor manufacturing industry.

TECHNICAL SKILLS

AI/LLM:	LangGraph, LangChain, LLaMA 3.2/3.3 70B, Groq API, Ollama, RAG, agentic systems, prompt engineering, SSE streaming
Inference & GPU:	RTX 3060 local inference, model quantisation awareness, Whisper (STT), Stable Diffusion
Vector Databases:	ChromaDB (production), FAISS (familiar); exploring Pinecone, Qdrant
Backend:	Python, Flask, REST APIs; FastAPI (in transition), MySQL, async patterns
Data & ML:	scikit-learn, Random Forest, pandas, numpy, OpenCV, NLP
Scraping & Ingestion:	Bright Data, automated web scraping pipelines, PDF extraction (ReportLab)
Other:	Unity 6 / C# / Meta Quest 3, Arduino C++, Raspberry Pi, ESP32, Git, MATLAB, SolidWorks

AI ENGINEERING PROJECTS

ARIA ? Agentic Biomedical Research Pipeline

- Developed a 4-stage LangGraph pipeline utilizing RTX 3060 for local inference, demonstrating expertise in high-performance computing and graphics processing
- Designed and implemented a real-time confidence scoring system using SSE streaming, showcasing ability to drive innovation in AI systems
- Built a fully automated pipeline for paper retrieval, evidence extraction, and auto-generated cited literature review, highlighting proficiency in agentic systems and prompt engineering

VERDICT ? Multi-Agent Candidate Evaluation System

- Orchestrated a parallel multi-agent pipeline evaluating candidates across three domains, leveraging LangGraph and LLaMA for AI-driven decision-making
- Developed a composite Stealth Signal Score metric to standardize evaluation output, demonstrating ability to drive growth through innovative solutions
- Built and submitted the project within the hackathon timeframe, targeting all three hackathon tracks, and published the full codebase on GitHub

UMER ? Local Personal AI Agent

- Designed and developed a fully local AI agent on RTX 3060, utilizing ChromaDB for persistent vector memory and Whisper for voice input
- Built a privacy-first architecture with all inference on-device, demonstrating expertise in high-performance computing and graphics processing
- Developed a smart query routing system and automated PowerPoint generation, highlighting ability to drive innovation in AI systems

PHANTOM ? Founder Intelligence Platform

- Built an MVP for a founder/startup signal scoring platform using automated GitHub and ProductHunt data ingestion
- Validated end-to-end flow locally, leveraging Flask and LangGraph for AI-driven decision-making
- Archived the project after MVP validation, demonstrating ability to drive growth through innovative solutions

ML Depression Screening

- Developed a keystroke-dynamics-based depression screening system using Python, scikit-learn, and Random Forest
- Achieved 87.5% accuracy and ROC-AUC 0.905, demonstrating expertise in machine learning and data analysis
- Deployed a live Streamlit app and prepared a manuscript for peer-reviewed submission

VR Trauma Surgery Simulator

- Designed and developed a real-time cardiovascular, respiratory, and nervous system state machines using Unity 6 and C#
- Built an interrupt-driven treatment logic mirroring embedded medical device architecture, demonstrating ability to drive innovation in AI systems
- Filed a provisional patent for the project, highlighting expertise in high-performance computing and graphics processing

WORK EXPERIENCE

Clinical Intern, CyberKnife Radiosurgery Unit, JPMC

June 2026 ? July 2026

- Rotated through the stereotactic radiosurgery unit, gaining hands-on exposure to clinical AI-adjacent workflows and high-precision medical device operation
- Developed understanding of the importance of high-performance computing and graphics processing in medical applications

EDUCATION

- BEng Biomedical Engineering, Ziauddin University, Karachi, 2024-2027 (6th Semester)